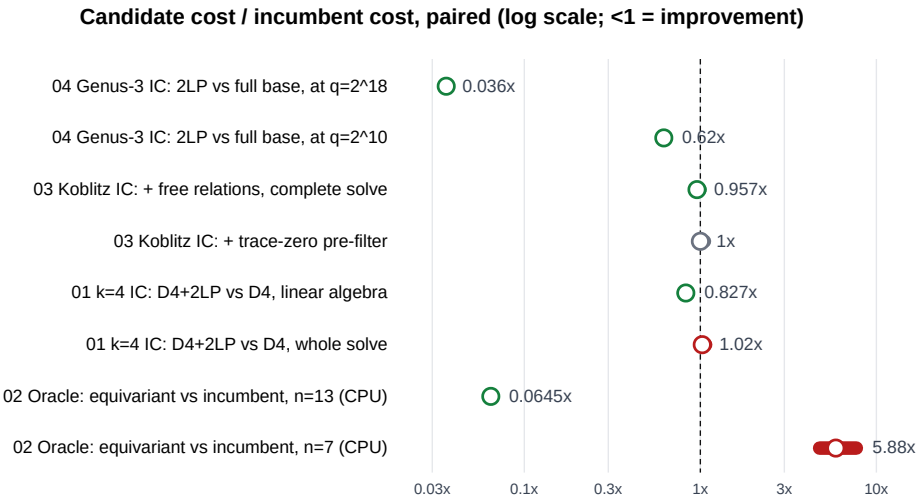
# IC improvement round

# IC improvement round: results (2026-10-14)

This round ran four follow-ups to [../ic_next10_20261014/README.md](../ic_next10_20261014/README.md), under [RULES.md](RULES.md):

- Each candidate is measured against the incumbent IC component it replaces.
- Every comparison is paired: same curves, targets and seeds, and the same counted unit on both arms.
- Every logarithm is verified.
- Rho is a reference column only; no verdict depends on it.




Exponent: genus-3 2LP q^1.385 vs incumbent q^1.910 (Δ −0.525); D4+2LP Δ +0.0055 (regress).
95% CIs. Green: improvement; red: regression; grey: no effect. Shared 4-core host, L0, counted units (item 02: CPU time).

Improvement over the incumbent

| # | Candidate vs incumbent | Verdict | Level | Key numbers |
|---|---|---|---|---|
| 04 | Genus-3 IC after cubic GHS: double large primes vs full factor base | **ADVANCES** | exponent and constant | |
| 03 | Compact-orbit Koblitz IC plus free relations harvested from the root table | **ADVANCES** | constant | |
| 03 | Same pipeline plus the trace-zero pre-filter | MATCHES / REGRESSES | primitive | |
| 01 | k=4 IC on E(F_{p⁴}): D₄ plus double large primes (tuned, frozen) vs D₄ | **TRADES**: whole cost REGRESSES | constant and exponent | |
| 02 | Koblitz decomposition oracle: G2 equivariant pass vs `solve_boolean_system` (InheritedF4 deg 3) | **TRADES** | constant | |

## Corrections to the earlier campaign

1. **Item 7's exponent comparison was not like for like.** Its "1.30 vs 1.65" compared arms over different size ranges. On a common range, the gain is larger: 1.385 vs 1.910, Δ −0.525. The candidate's exponent keeps rising: its local exponent reaches 1.46 at 2^20. The cause is the linear algebra, which scales as q^1.52 because row weight grows after large-prime elimination. So the exponent does not settle at the theoretical 4/3 with this linear algebra and κ = 2.
2. **Item 6's 44–73× equivariant speedup was measured against a naive elimination kernel.** Against the crate's `gf2_elim` it is 2–3× at n = 11–13.
3. **Item 8's trace-zero "50% refutation" does not apply to IC targets,** which have odd order.

## Where to push next (improvement targets)

- **Genus-3 linear algebra.** It is now the bottleneck: 65% of cost at 2^20, scaling as q^1.52. Two options:

    - a structured-Gaussian-elimination prefilter, or block Wiedemann on the sparse matrix, instead of Lanczos on the normal equations;
    - re-tuning κ at large q.

  Either should pull the exponent toward 4/3.
- **Free relations in the Koblitz pipeline.** Pick K so the harvested rank is a large fraction, as at n = 41 (63%), and land the change behind a flag in the production example.
- **Fall-row augmentation (item 02, arm F).** It gave 11.6× on barrel systems but regresses on pipeline systems. A cheap per-system predictor of when it helps would be needed before it is worth wiring in.

## Caveats

- **Host:** shared, 4 cores, load average up to about 25, isolation level L0. Operation counts are the measured unit.
- **Small samples:** 3 curves per size in item 04. Its A/A check shows a ±2% noise floor on per-size ratio intervals.
- **Python:** used for figures only.
- **Conductor:** the control plane was unreachable.